

Interpretation of fixed effects in linear mixed models and generalized linear mixed models

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Clustered/longitudinal data: outcome Y_{ij} , and covariates $\mathbf{X}_{ij} = (gender_i, age_{ij}, bmi_{ij})$ with level 2 index i and level 1 index j .

1 Normal outcome

Look at a multilevel model with random intercept,

$$Y_{ij} = \beta_i + \beta_1 gender_i + \beta_2 age_{ij} + \beta_3 bmi_{ij} + \epsilon_{ij}, \quad \beta_i = \beta_0 + \gamma_i,$$

where $\gamma_i \stackrel{i.i.d}{\sim} N(0, \sigma_\gamma^2)$, $\epsilon_{ij} \stackrel{i.i.d}{\sim} N(0, \sigma_0^2)$ and γ_i is independent of ϵ_{ij} .

The conditional expectation,

$$E(Y_{ij} | \beta_i, gender_i, age_{ij}, bmi_{ij}) = \beta_i + \beta_1 gender_i + \beta_2 age_{ij} + \beta_3 bmi_{ij},$$

and the marginal expectation

$$E(Y_{ij} | gender_i, age_{ij}, bmi_{ij}) = \beta_0 + \beta_1 gender_i + \beta_2 age_{ij} + \beta_3 bmi_{ij}.$$

- The interpretation of any component of β , say β_2 , is in terms of changes in the mean response for a unit change in the corresponding covariate, say age_{ij}
- When age_{ij} takes on some value x , the mean of the response is

$$E(Y_{ij} | gender_i, age_{ij} = x, bmi_{ij}) = \beta_0 + \beta_1 gender_i + \beta_2 x + \beta_3 bmi_{ij}.$$

- Similarly, when age_{ij} now takes on the value $x + 1$,

$$E(Y_{ij} | gender_i, age_{ij} = x + 1, bmi_{ij}) = \beta_0 + \beta_1 gender_i + \beta_2(x + 1) + \beta_3 bmi_{ij}.$$

- β_2 is the change in the mean response for **subgroups of the population**, defined by any fixed values of $gender_i$ and bmi_{ij} , e.g., men with bmi 22.

$$\beta_2 = E(Y_{ij} | gender_i, age_{ij} = x + 1, bmi_{ij}) - E(Y_{ij} | gender_i, age_{ij} = x, bmi_{ij}).$$

2 Non-normal outcome

For review, consider the ordinary logistic regression,

$$\text{logit}E(Y_{ij}|\mathbf{X}_{ij}) = \text{logit} \frac{\Pr(Y_{ij} = 1|\mathbf{X}_{ij})}{\Pr(Y_{ij} = 0|\mathbf{X}_{ij})} = \beta_0 + \beta_1 \text{gender}_i + \beta_2 \text{age}_{ij} + \beta_3 \text{bmi}_{ij}$$

- Each element of β measures the change in the log odds of a “positive” response (success) per unit change in the respective covariate, for subpopulations defined by fixed and known covariate values
- The interpretation of any component of β , say β_2 , is in terms of changes in the transformed mean response for a unit change in the corresponding covariate, say age_{ij}
- When age_{ij} takes on some value x , the log odds of a positive response is

$$\log \left[\frac{\Pr(Y_{ij}=1|\text{gender}_i, \text{age}_{ij} = x, \text{bmi}_{ij})}{\Pr(Y_{ij}=0|\text{gender}_i, \text{age}_{ij} = x, \text{bmi}_{ij})} \right] = \beta_0 + \beta_1 \text{gender}_i + \beta_2 x + \beta_3 \text{bmi}_{ij}.$$

- Similarly, when age_{ij} now takes on the value $x + 1$,

$$\log \left[\frac{\Pr(Y_{ij}=1|\text{gender}_i, \text{age}_{ij} = x + 1, \text{bmi}_{ij})}{\Pr(Y_{ij}=0|\text{gender}_i, \text{age}_{ij} = x + 1, \text{bmi}_{ij})} \right] = \beta_0 + \beta_1 \text{gender}_i + \beta_2 (x + 1) + \beta_3 \text{bmi}_{ij}.$$

- β_2 is the change in log odds for **subgroups of the population**, defined by any fixed values of gender_i and bmi_{ij} , e.g., men with bmi 22.

$$\beta_2 = \log \left[\frac{\Pr(Y_{ij}=1|\text{gender}_i, \text{age}_{ij} = x + 1, \text{bmi}_{ij})}{\Pr(Y_{ij}=0|\text{gender}_i, \text{age}_{ij} = x + 1, \text{bmi}_{ij})} \right] - \log \left[\frac{\Pr(Y_{ij}=1|\text{gender}_i, \text{age}_{ij} = x, \text{bmi}_{ij})}{\Pr(Y_{ij}=0|\text{gender}_i, \text{age}_{ij} = x, \text{bmi}_{ij})} \right].$$

However, this is different from the interpretation of regression coefficients in generalized linear mixed models, where the coefficients of fixed effects measure the change on the **individual level** in the expected values of response while holding other covariates and the random effects as constant. Consider a logistic regression with random intercept,

$$\text{logit}E(Y_{ij}|\beta_i, \mathbf{X}_{ij}) = \text{logit} \frac{\Pr(Y_{ij} = 1|\beta_i, \mathbf{X}_{ij})}{\Pr(Y_{ij} = 0|\beta_i, \mathbf{X}_{ij})} = \beta_i + \beta_1 \text{gender}_i + \beta_2 \text{age}_{ij} + \beta_3 \text{bmi}_{ij}$$

- Each element of fixed effects β measures the change in the log odds of a “positive” response (success) per unit change in the respective covariate, for one particular individual
- The interpretation of any component of β , say β_2 , is in terms of changes in the transformed mean response for a unit change in the corresponding covariate, say age_{ij} , for individual i
- When age_{ij} takes on some value x , the log odds of a positive response is

$$\log \left[\frac{\Pr(Y_{ij}=1|\beta_i, \text{gender}_i, \text{age}_{ij} = x, \text{bmi}_{ij})}{\Pr(Y_{ij}=0|\beta_i, \text{gender}_i, \text{age}_{ij} = x, \text{bmi}_{ij})} \right] = \beta_i + \beta_1 \text{gender}_i + \beta_2 x + \beta_3 \text{bmi}_{ij}.$$

- Similarly, when age_{ij} now takes on the value $x + 1$,

$$\log\left[\frac{Pr(Y_{ij=1}|\beta_i, gender_i, age_{ij} = x + 1, bmi_{ij})}{Pr(Y_{ij=0}|\beta_i, gender_i, age_{ij} = x + 1, bmi_{ij})}\right] = \beta_i + \beta_1 gender_i + \beta_2(x + 1) + \beta_3 bmi_{ij}.$$

- β_2 is the change in log odds for individual i .

$$\beta_2 = \log\left[\frac{Pr(Y_{ij=1}|\beta_i, gender_i, age_{ij} = x + 1, bmi_{ij})}{Pr(Y_{ij=0}|\beta_i, gender_i, age_{ij} = x + 1, bmi_{ij})}\right] - \log\left[\frac{Pr(Y_{ij=1}|\beta_i, gender_i, age_{ij} = x, bmi_{ij})}{Pr(Y_{ij=0}|\beta_i, gender_i, age_{ij} = x, bmi_{ij})}\right].$$